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Studierichting: Informatica
Oefeningenreeks 4A nr. 45.1

Opgave: $I = \int \frac{9x^2 - x + 2}{4x^5 + x^3} dx$

$= \int \frac{9x^2 - x + 2}{x^3 \cdot (4x^2 + 1)} dx$

$\Rightarrow \frac{9x^2 - x + 2}{(4x^2 + 1) \cdot x^3} = \frac{Ax + B}{4x^2 + 1} + \frac{C}{x} + \frac{D}{x^2} + \frac{E}{x^3}$

$\Rightarrow 4x^2 + 1 = 0 \Rightarrow x = \pm \frac{1}{2} i$

$$A\left(\frac{1}{2}\right) + B = \left.\frac{9x^2 - x + 2}{x^3}\right|_{x=\frac{1}{2}i} = \frac{9\left(\frac{1}{2}i\right)^2 - \left(\frac{1}{2}i\right) + 2}{\left(\frac{1}{2}i\right)^3}$$

$$= \frac{-\frac{9}{4} - \frac{2}{4}i + \frac{8}{4}}{-\frac{1}{8}i} = \frac{\frac{1}{4} + \frac{2}{4}i}{\frac{1}{8}i} = \frac{8(1+2i)}{4i} = \frac{2+4i}{i}\frac{i}{i} = \frac{2i-4}{-1} = 4 - 2i$$

$$\Rightarrow \left\{ \begin{array}{lcr} A\left(\frac{1}{2}i\right) & = & -2i \\ B & = & 4 \end{array} \right. \Rightarrow (A, B) = (-4, 4)$$

$$E = \left.\frac{9x^2 - x + 2}{4x^2 + 1}\right|_{x=0} = \frac{9 \cdot 0 - 0 + 2}{4 \cdot 0 + 1} = 2$$

$$\Rightarrow \frac{9x^2 - x + 2}{(4x^2 + 1) \cdot x^3} - \frac{2}{x^3} = \frac{9x^2 - x + 2 - 8x^2 - 2}{(4x^2 + 1) \cdot x^3} = \frac{x^2 - x}{(4x^2 + 1) \cdot x^3} = \frac{x - 1}{(4x^2 + 1) \cdot x^2}$$

$$\Rightarrow D = \left.\frac{x - 1}{4x^2 + 1}\right|_{x=0} = \frac{0 - 1}{4 \cdot 0 + 1} = -1$$

$$\Rightarrow \frac{x - 1}{(4x^2 + 1) \cdot x^2} - \left(-\frac{1 \cdot (4x^2 + 1)}{x^2 \cdot (4x^2 + 1)}\right) = \frac{x - 1 + x^2 + 1}{x^2 \cdot (4x^2 + 1)} = \frac{x^2 + x}{x^2 \cdot (4x^2 + 1)}$$

$$= \frac{x + 1}{x \cdot (4x^2 + 1)}$$

$$\Rightarrow C = \left.\frac{x + 1}{4x^2 + 1}\right|_{x=0} = \frac{0 + 1}{4 \cdot 0 + 1} = 1$$

$$I = \int \frac{dx}{x} - \int \frac{dx}{x^2} + 2\int \frac{dx}{x^3} + \int \frac{-4x + 4}{4x^2 + 1} dx$$

$$\Rightarrow -4 \int \frac{x - 1}{4x^2 + 1} dx = -4 \left(\int \frac{x dx}{4x^2 + 1} - \int \frac{dx}{(2x)^2 + 1} \right) = -4 \left(\frac{1}{8} \int \frac{d(4x^2 + 1)}{4x^2 + 1} - \frac{1}{2} \int \frac{d(2x)}{(2x)^2 + 1} \right)$$

$$= -\frac{1}{2} \ln |4x^2 + 1| + 2 \operatorname{Bgtan}(2x) + c$$

$$\Rightarrow I = \ln |x| - \frac{x^{-1}}{-1} + 2\frac{x^{-2}}{-2} - \frac{1}{2} \ln(4x^2 + 1) + 2 \operatorname{Bgtan}(2x) + c$$

$$= \ln|x| + \frac{1}{x} - \frac{1}{x^2} - \frac{1}{2}\ln(4x^2+1) + 2\,\mathrm{Bgtan}(2x) + c$$

References